

RAI 818 – INTELLIGENT MOBILE ROBOTICS

Mobile Robot Kinematics - II

- **Mobile Robot Kinematics Using ICC**
- **Constraints**

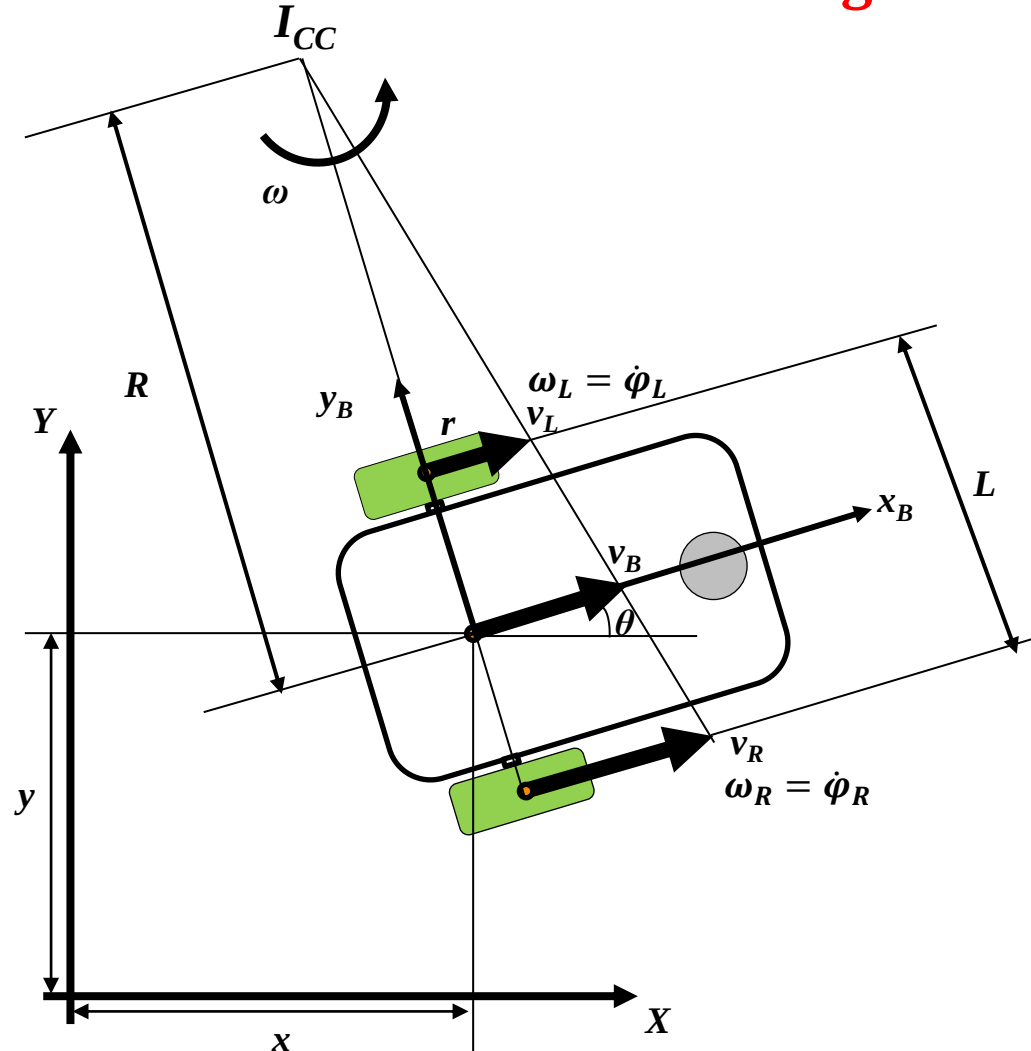
Text : Principles of Robot Motion – Theory, Algorithms and Implementations
H. Choset, K.M. Lynch, S. Hutchinson, G. Kantor, W. Burgard, et al.

INSTRUCTOR:
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Kinematics and Dynamics

- Kinematics
 - Mobile Robot Kinematics Using ICC
 - Differential Drive

Mobile Robot Kinematics Using ICC – Differential Drive



$$v_B = R\omega$$

$$v_L = \omega \left(R - \frac{L}{2} \right) \quad v_R = \omega \left(R + \frac{L}{2} \right)$$

$$\omega = \frac{v_L}{\left(R - \frac{L}{2} \right)} \quad v_R = \frac{v_L}{\left(R - \frac{L}{2} \right)} \left(R + \frac{L}{2} \right)$$

$$v_R \left(R - \frac{L}{2} \right) = v_L \left(R + \frac{L}{2} \right)$$

$$R(v_R - v_L) = v_R \frac{L}{2} + v_L \frac{L}{2}$$

$$R = \frac{L(v_R + v_L)}{2(v_R - v_L)}$$

$$\omega = \frac{v_R - v_L}{L} \quad \omega = \frac{v_R + v_L}{2R}$$



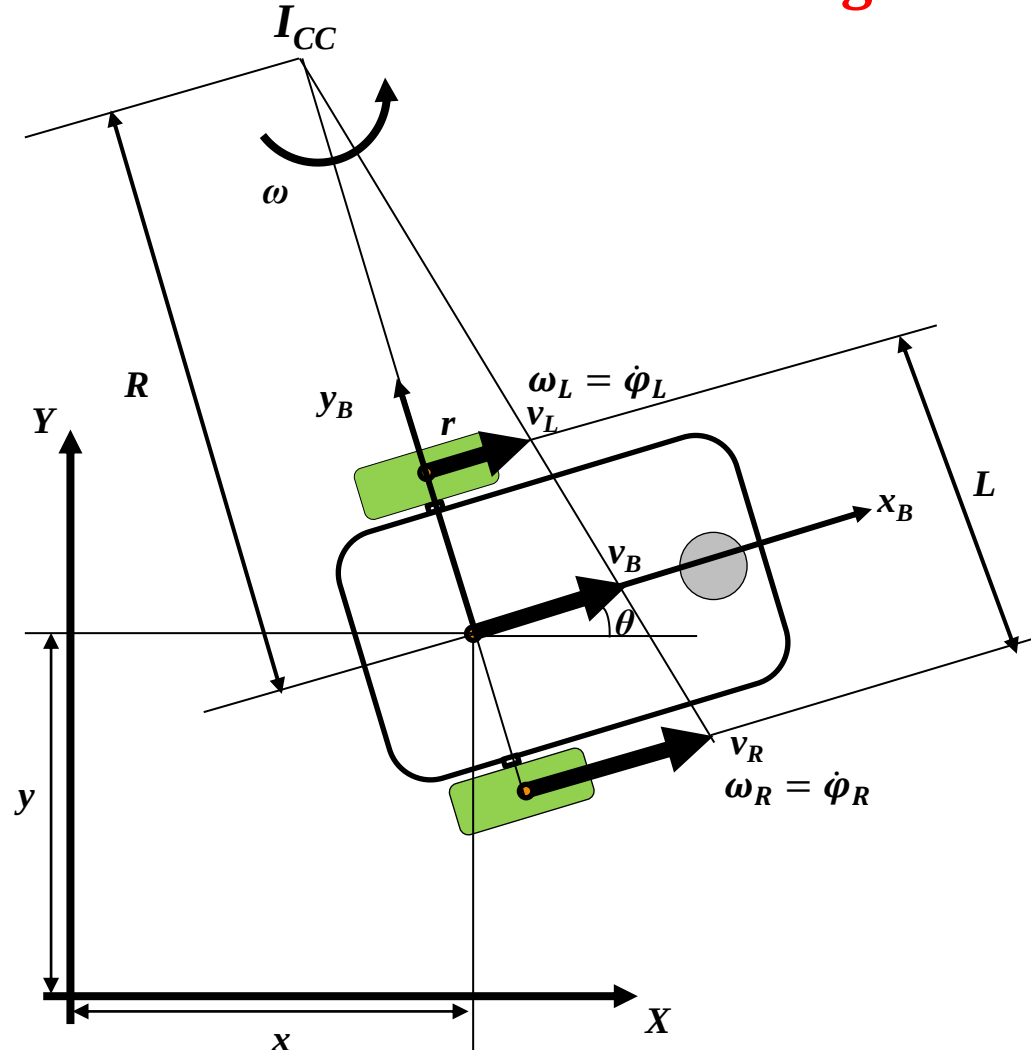
Kinematics and Dynamics



R&AI

- Kinematics
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Mobile Robot Kinematics Using ICC – Differential Drive



$$\begin{aligned} \dot{x} &= v_B \cos \theta \\ \dot{y} &= v_B \sin \theta \\ \dot{\theta} &= \omega \end{aligned} \quad \begin{bmatrix} \dot{x} \\ \dot{y} \\ \dot{\theta} \end{bmatrix} = \begin{bmatrix} \cos \theta & 0 \\ \sin \theta & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} v_B \\ \omega \end{bmatrix}$$

$$\begin{aligned} v_B &= \frac{v_R - v_L}{L} \cdot \frac{L(v_R + v_L)}{2(v_R - v_L)} \\ v_B &= \frac{v_R + v_L}{2} \quad \omega = \frac{v_R - v_L}{L} \end{aligned} \quad \begin{bmatrix} v_B \\ \omega \end{bmatrix} = \begin{bmatrix} \frac{1}{2} & \frac{1}{2} \\ -\frac{1}{L} & \frac{1}{L} \end{bmatrix} \begin{bmatrix} v_L \\ v_R \end{bmatrix}$$

$$\begin{bmatrix} \dot{x} \\ \dot{y} \\ \dot{\theta} \end{bmatrix} = \begin{bmatrix} \cos \theta & 0 \\ \sin \theta & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} \frac{1}{2} & \frac{1}{2} \\ -\frac{1}{L} & \frac{1}{L} \end{bmatrix} \begin{bmatrix} v_L \\ v_R \end{bmatrix} = \begin{bmatrix} \frac{\cos \theta}{2} & \frac{\cos \theta}{2} \\ \frac{\sin \theta}{2} & \frac{\sin \theta}{2} \\ -\frac{1}{L} & \frac{1}{L} \end{bmatrix} \begin{bmatrix} v_L \\ v_R \end{bmatrix}$$

$$\dot{x} = \frac{v_L}{2} \cos \theta + \frac{v_R}{2} \cos \theta$$

$$\dot{y} = \frac{v_L}{2} \sin \theta + \frac{v_R}{2} \sin \theta$$

$$\dot{\theta} = -\frac{1}{L} v_L + \frac{1}{L} v_R$$

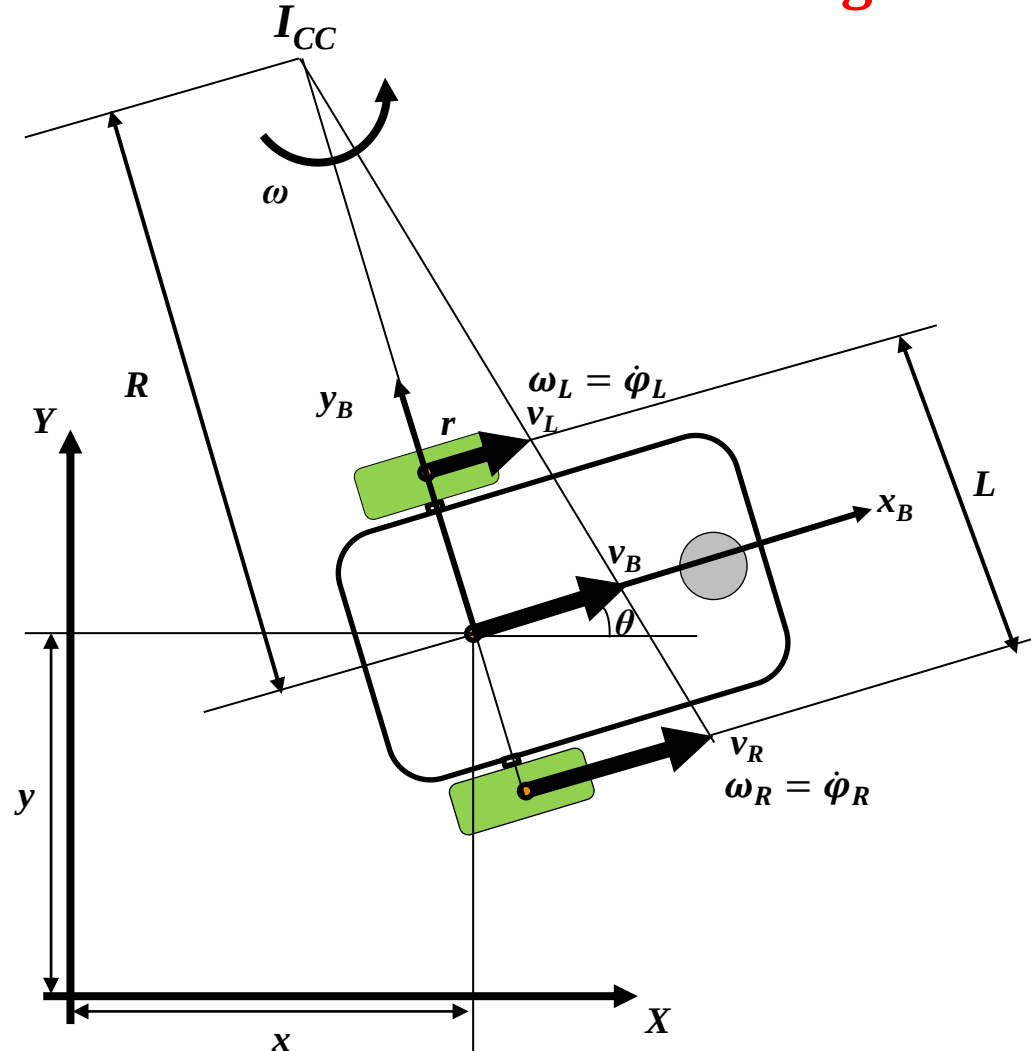


Defining futures

Kinematics and Dynamics

- Kinematics
 - Mobile Robot Kinematics Using ICC
 - Differential Drive

Mobile Robot Kinematics Using ICC – Differential Drive



$$\begin{bmatrix} \dot{x} \\ \dot{y} \\ \dot{\theta} \end{bmatrix} = \begin{bmatrix} \frac{\cos \theta}{2} & \frac{\cos \theta}{2} \\ \frac{\sin \theta}{2} & \frac{\sin \theta}{2} \\ \frac{1}{L} & -\frac{1}{L} \end{bmatrix} \begin{bmatrix} v_L \\ v_R \end{bmatrix}$$

$$v_L = r\omega_L \quad v_R = r\omega_R$$

$$\begin{bmatrix} v_L \\ v_R \end{bmatrix} = \begin{bmatrix} r & 0 \\ 0 & r \end{bmatrix} \begin{bmatrix} \omega_L \\ \omega_R \end{bmatrix}$$

$$\begin{bmatrix} \dot{x} \\ \dot{y} \\ \dot{\theta} \end{bmatrix} = \begin{bmatrix} \frac{\cos \theta}{2} & \frac{\cos \theta}{2} \\ \frac{\sin \theta}{2} & \frac{\sin \theta}{2} \\ \frac{1}{L} & -\frac{1}{L} \end{bmatrix} \begin{bmatrix} r & 0 \\ 0 & r \end{bmatrix} \begin{bmatrix} \omega_L \\ \omega_R \end{bmatrix} = \begin{bmatrix} \frac{r \cos \theta}{2} & \frac{r \cos \theta}{2} \\ \frac{r \sin \theta}{2} & \frac{r \sin \theta}{2} \\ -\frac{r}{L} & \frac{r}{L} \end{bmatrix} \begin{bmatrix} \omega_L \\ \omega_R \end{bmatrix}$$

$$\dot{x} = \frac{r\omega_L}{2} \cos \theta + \frac{r\omega_R}{2} \cos \theta$$

$$\dot{y} = \frac{r\omega_L}{2} \sin \theta + \frac{r\omega_R}{2} \sin \theta$$

$$\dot{\theta} = -\frac{r}{L} \omega_L + \frac{r}{L} \omega_R$$

Kinematics and Dynamics

- Kinematics
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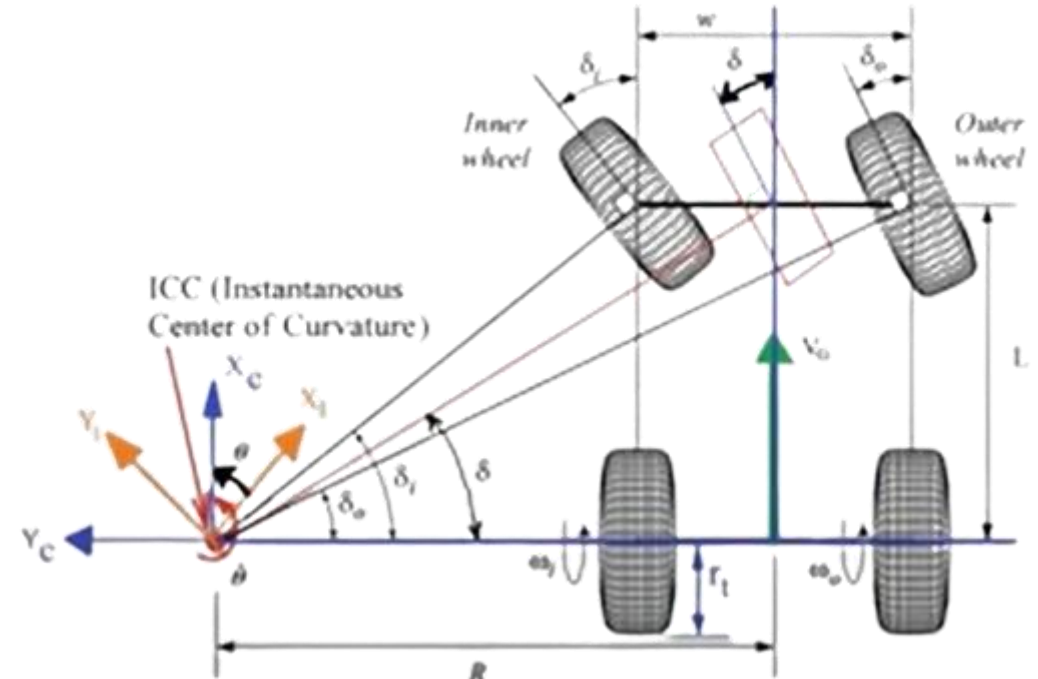
Mobile Robot Kinematics Using ICC – Differential Drive

$$\tan(\delta_i) = \frac{L}{R - \frac{w}{2}} \quad \tan(\delta_o) = \frac{L}{R + \frac{w}{2}}$$

$$\Rightarrow \cot(\delta_o) - \cot(\delta_i) = \frac{R + \frac{w}{2}}{L} - \frac{R - \frac{w}{2}}{L} = \frac{w}{L}$$

$$\tan(\delta) = \frac{L}{R}$$

$$\gamma = \frac{\delta_i}{\delta} : \text{steering ratio}$$



w is called the track width
L is called the wheel base

$$\Rightarrow \cot(\delta_o) - \cot(\delta) = \frac{R + \frac{w}{2}}{L} - \frac{R}{L} = \frac{w}{2L} \Rightarrow \delta_o = \tan^{-1} \left(\frac{L \times \tan \delta}{L - 0.5 \times w \times \tan \delta} \right) \quad (2)$$

$$\Rightarrow \cot(\delta) - \cot(\delta_i) = \frac{R}{L} - \frac{R - \frac{w}{2}}{L} = \frac{w}{2L} \Rightarrow \delta_i = \tan^{-1} \left(\frac{L \times \tan \delta}{L + 0.5 \times w \times \tan \delta} \right) \quad (3)$$



Kinematics and Dynamics

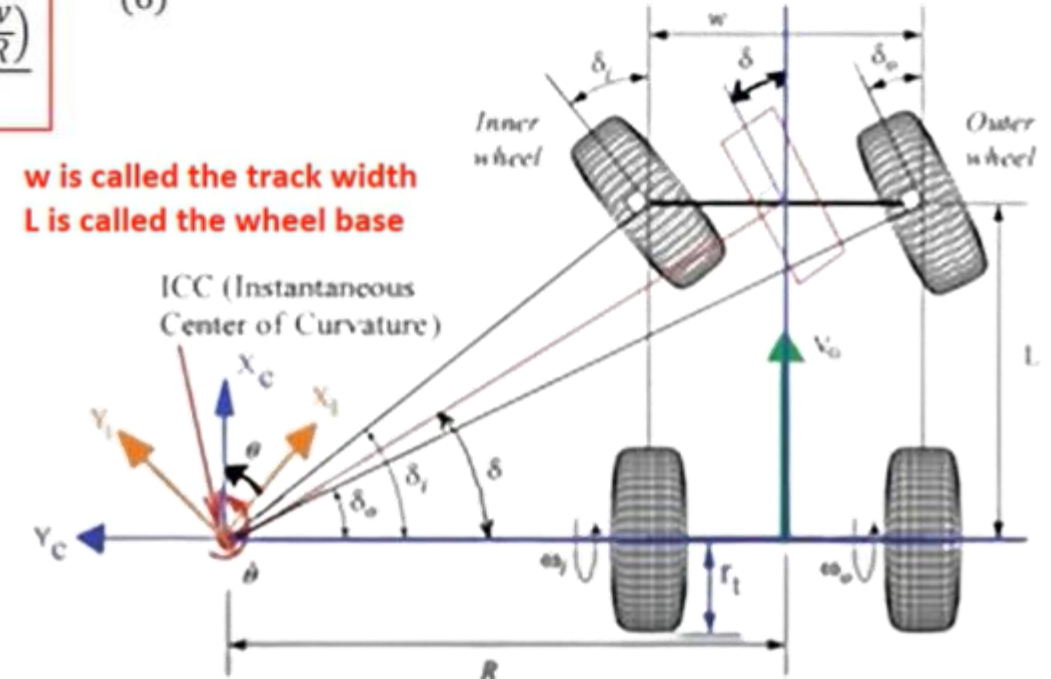
- Kinematics
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Mobile Robot Kinematics Using ICC – Differential Drive

$$\begin{cases} \left(R + \frac{w}{2}\right) \dot{\theta} = V_{outer_wheel_center} = r_t \omega_o \\ \left(R - \frac{w}{2}\right) \dot{\theta} = V_{inner_wheel_center} = r_t \omega_i \end{cases} \Rightarrow \begin{cases} w\dot{\theta} = r_t \omega_o - r_t \omega_i & (4) \\ 2R\dot{\theta} = 2V_G = r_t \omega_o + r_t \omega_i \Rightarrow V_G = \frac{r_t \omega_o + r_t \omega_i}{2} & (5) \end{cases}$$

$$(4), (5) \Rightarrow \begin{cases} \frac{wV_G}{R} = r_t(\omega_o - \omega_i) \\ 2V_G = r_t \omega_o + r_t \omega_i \end{cases} \Rightarrow \begin{cases} \omega_o = \frac{V_G \left(2 + \frac{w}{R}\right)}{2r_t} \\ \omega_i = \frac{V_G \left(2 - \frac{w}{R}\right)}{2r_t} \end{cases} \quad (6)$$

$$\begin{cases} \dot{X} = V_G \cos \theta \\ \dot{Y} = V_G \sin \theta \\ \dot{\theta} = \frac{V_G}{R} = \frac{V_G}{\frac{L}{\tan(\delta)}} = \frac{V_G \tan(\delta)}{L} \end{cases} \quad (7)$$



Kinematics and Dynamics

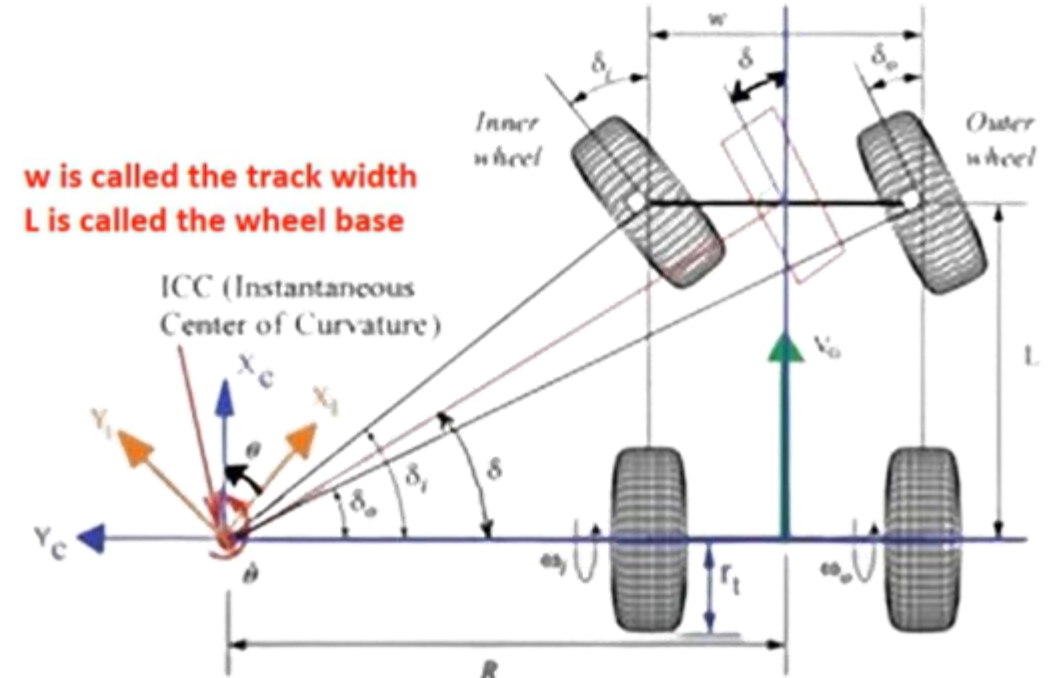
- Kinematics
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Mobile Robot Kinematics Using ICC – Differential Drive

$$\begin{cases} \dot{X} = V_G \cos \theta = V_G \cos \left(\frac{V_G}{R} t + \theta_0 \right) \Rightarrow X = R \sin \left(\frac{V_G}{R} t + \theta_0 \right) + X_0 - R \sin(\theta_0) \\ \dot{Y} = V_G \sin \theta = V_G \sin \left(\frac{V_G}{R} t + \theta_0 \right) \Rightarrow Y = -R \cos \left(\frac{V_G}{R} t + \theta_0 \right) + R \cos(\theta_0) + Y_0 \\ \text{if } \tan(\delta) = \frac{L}{R} = \text{const} \Rightarrow \dot{\theta} = \frac{V_G}{R} = \text{const} \Rightarrow \theta = \frac{V_G}{R} t + \theta_0 \end{cases}$$

$$\Rightarrow (X - X_0 + R \sin(\theta_0))^2 + (Y - R \cos(\theta_0) - Y_0)^2 = R^2$$

$$\Rightarrow T = \frac{2\pi}{\omega} = \frac{2\pi}{\frac{V_G}{R}} = \frac{2\pi R}{V_G}$$



Kinematics and Dynamics



- Kinematics
 - Mobile Robot Kinematics Using ICC
 - Constraints

Constraints

- Geometric constraints are generally not sufficient to adequately express robot motions
- Constraints on velocity, forces, torques, accelerations are needed for better representations

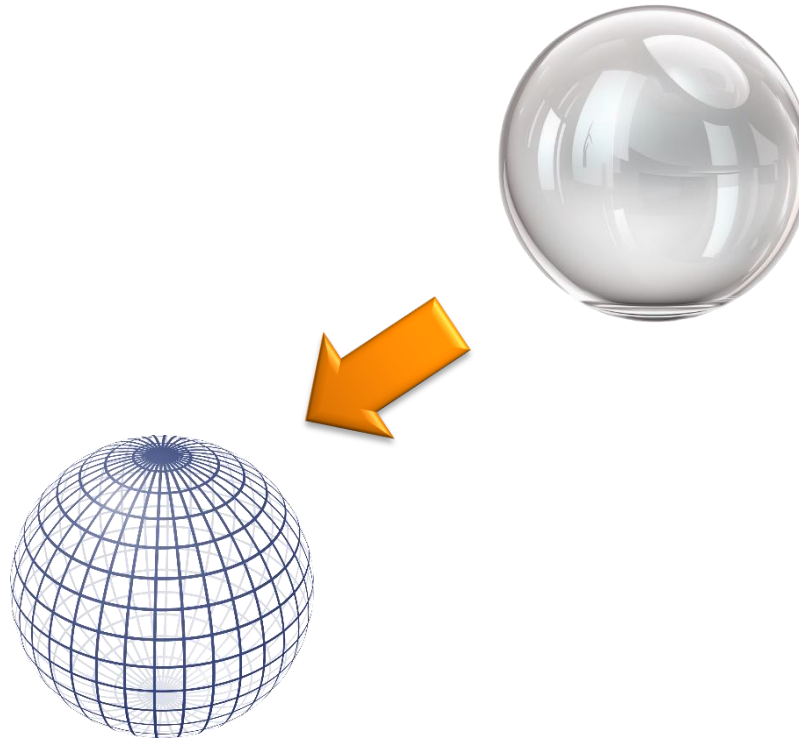


Defining futures

Kinematics and Dynamics

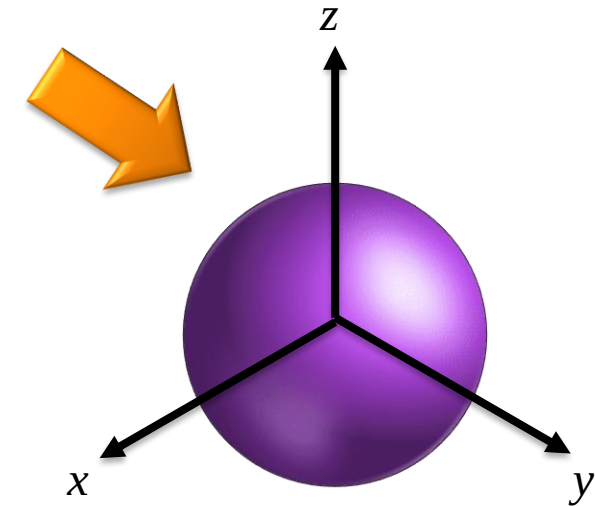
- Kinematics
 - Mobile Robot Kinematics Using ICC
 - Constraints

Constraints



Explicit Parameterization

Minimum Number of Coordinates
E.g., Longitude, Latitude



Implicit Representation

A surface embedded in a higher dimensional space
E.g., x, y, z
 $x^2 + y^2 + z^2 = 1$



Kinematics and Dynamics

- Kinematics
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Constraints

$$L_1 \cos(\theta_1) + L_2 \cos(\theta_1 + \theta_2) + L_3 \cos(\theta_1 + \theta_2 + \theta_3) + L_4 \cos(\theta_1 + \theta_2 + \theta_3 + \theta_4) = 0$$

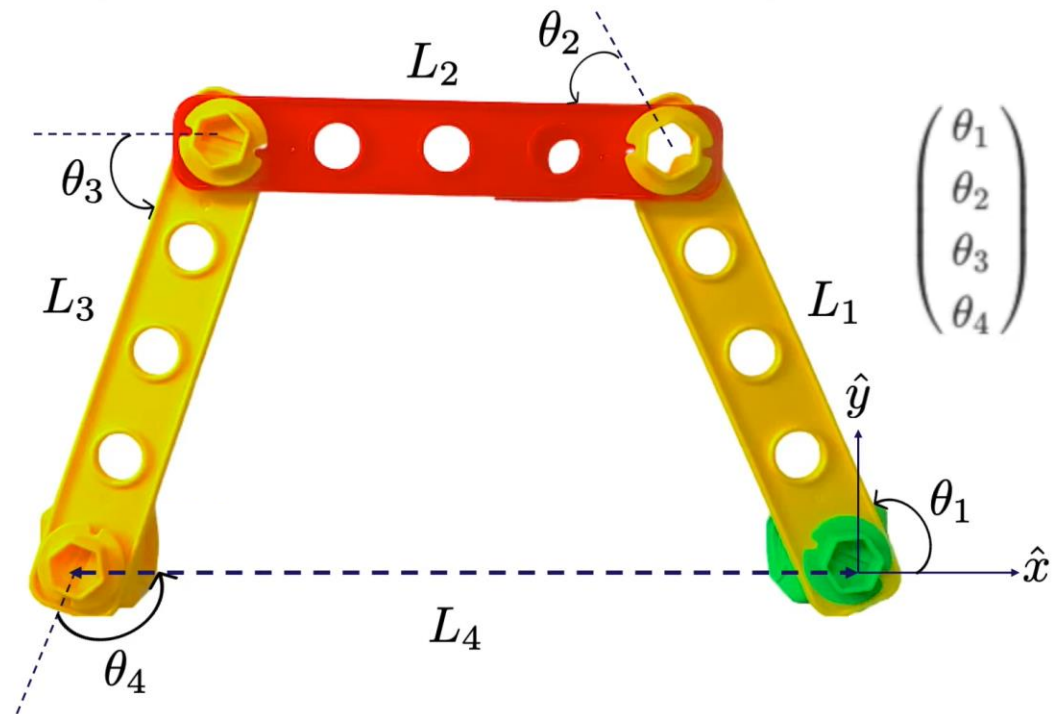
$$L_1 \sin(\theta_1) + L_2 \sin(\theta_1 + \theta_2) + L_3 \sin(\theta_1 + \theta_2 + \theta_3) + L_4 \sin(\theta_1 + \theta_2 + \theta_3 + \theta_4) = 0$$

$$\theta_1 + \theta_2 + \theta_3 + \theta_4 - 2\pi = 0$$

$$q = [q_1 \quad \cdot \quad \cdot \quad \cdot \quad q_n]^T \in R^n$$

$$g(q) = \begin{bmatrix} g_1(q_1 \quad \cdot \quad \cdot \quad \cdot \quad q_n) \\ \cdot \\ \cdot \\ \cdot \\ g_k(q_1 \quad \cdot \quad \cdot \quad \cdot \quad q_n) \end{bmatrix} = 0 \quad k \leq n$$

$$g(q) \in R^k$$



Kinematics and Dynamics

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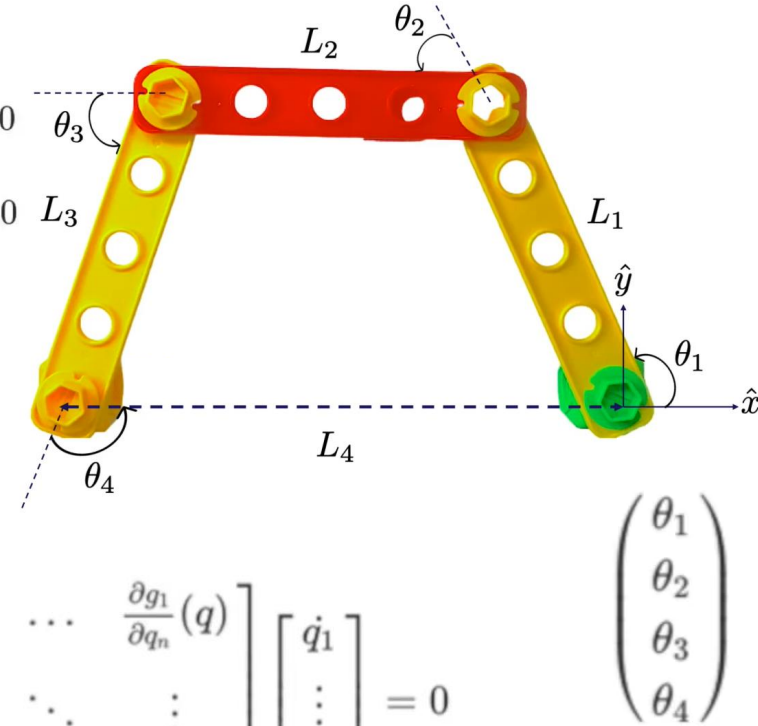
Constraints

$$L_1 \cos(\theta_1) + L_2 \cos(\theta_1 + \theta_2) + L_3 \cos(\theta_1 + \theta_2 + \theta_3) + L_4 \cos(\theta_1 + \theta_2 + \theta_3 + \theta_4) = 0$$

$$L_1 \sin(\theta_1) + L_2 \sin(\theta_1 + \theta_2) + L_3 \sin(\theta_1 + \theta_2 + \theta_3) + L_4 \sin(\theta_1 + \theta_2 + \theta_3 + \theta_4) = 0$$

$$\theta_1 + \theta_2 + \theta_3 + \theta_4 - 2\pi = 0$$

$$g(q(t)) = 0 \quad g: R^n \rightarrow R^k$$



$$\frac{d}{dt}g(q(t)) = 0 \rightarrow \begin{bmatrix} \frac{\partial g_1}{\partial q_1}(q)\dot{q}_1 + \dots + \frac{\partial g_1}{\partial q_n}(q)\dot{q}_n \\ \vdots \\ \frac{\partial g_k}{\partial q_1}(q)\dot{q}_1 + \dots + \frac{\partial g_k}{\partial q_n}(q)\dot{q}_n \end{bmatrix} = 0 \rightarrow \underbrace{\begin{bmatrix} \frac{\partial g_1}{\partial q_1}(q) & \dots & \frac{\partial g_1}{\partial q_n}(q) \\ \vdots & \ddots & \vdots \\ \frac{\partial g_k}{\partial q_1}(q) & \dots & \frac{\partial g_k}{\partial q_n}(q) \end{bmatrix}}_{A(q) = \frac{\partial g}{\partial q}(q) \in R^{k \times n}} \underbrace{\begin{bmatrix} \dot{q}_1 \\ \vdots \\ \dot{q}_n \end{bmatrix}}_{\dot{q} \in R^n} = 0$$

Pfaffian Constraints

$$A(q)\dot{q} = 0$$



Defining futures

- Introduction
- Coordinate Systems
- Kinematic Models
- Constraints
 - Implicit Velocity Constraints

Implicit Velocity Constraints

- Implicit velocity constraints express velocities that are not allowed, and are of the form $g(q, \dot{q}) \otimes 0$
where
 - $g(q, \dot{q})$ is some function $g : Q \times \dot{Q} \rightarrow \mathbb{R}$
 - $\otimes$ can be any of the symbols $=, <, >, \geq, \leq$
- Example of point in plane
 - configuration: $q = (x, y) \in \mathbb{R}^2$
 - velocity: $\frac{dq}{dt} = \dot{q} = (\dot{x}, \dot{y})$
- Examples of implicit velocity constraints
 - $\dot{x} = 0$
 - $\dot{x} > 0$
 - $\dot{x}^2 + \dot{y}^2 \leq 1$
 - $x = \dot{x}$



- Introduction
- Coordinate Systems
- Kinematic Models
- **Constraints**
 - Implicit Vel Constraints
 - **Parametric Vel Constraints**

Parametric Velocity Constraints

- Parametric velocity constraints express velocities that are allowed, and are of the form

where $\dot{q} = f(q, u)$

- $f(q, u)$ is some function $f : Q \times U \rightarrow \dot{Q}$ that expresses a set of differential equations
- u is an input control



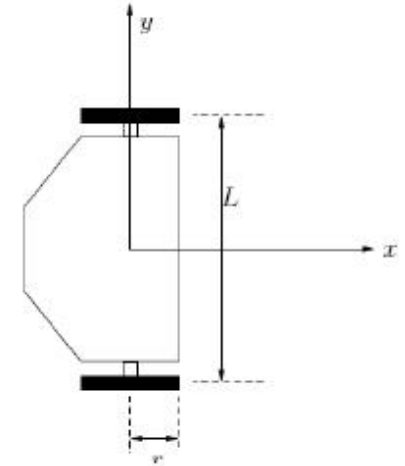
Defining futures

Kinematics and Dynamics

- Introduction
- Coordinate Systems
- Kinematic Models
- Constraints
- Applications
 - A Simple Car
 - Wheeled Robot

Kinematics for Wheeled Systems – Differential Drive

- *Input controls* $u = (u_l, u_r)$
 - u_l : angular velocity - left wheel
 - u_r : angular velocity- right wheel
- *How does the robot move?*
 - $u_l = u_r \Rightarrow$ moves forward in the direction the wheels are pointing speed proportional to wheel radius r
 - $u_l = -u_r$ rotates clockwise because wheels are turning in opposite directions
- *Where is the body frame placed?*
 - origin at the centre of the axle between the wheels then
$$\dot{x} = ru_\omega \cos \theta$$
$$\dot{y} = ru_\omega \sin \theta$$
$$\dot{\theta} = ru_\psi / L$$



Equations of Motions

$$\dot{x} = \frac{r}{2}(u_l + u_r) \cos \theta$$

$$\dot{y} = \frac{r}{2}(u_l + u_r) \sin \theta$$

$$\dot{\theta} = \frac{r}{L}(u_r - u_l)$$

rotate

$$u_\omega = \frac{(u_l + u_r)}{2}$$

translate

$$u_\phi = \frac{(u_r - u_l)}{2}$$

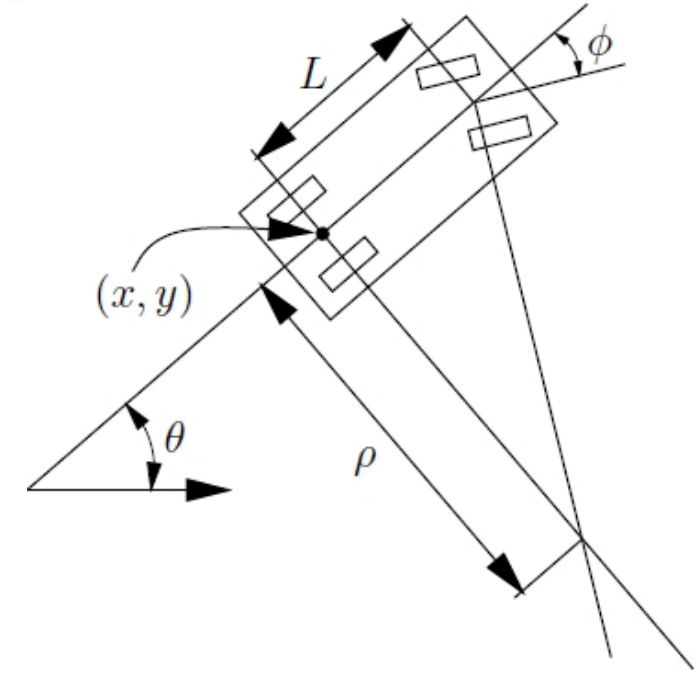
Can it move between any two configurations?



- Introduction
- Coordinate Systems
- Kinematic Models
- Constraints
- Applications
 - A Simple Car

Kinematics for Wheeled Systems – A Simple Car

- *Car config*: $q = (x, y, \theta) \in \mathbb{R} \times S_1$
- Body frame
 - *Origin*: centre of rear axle
 - *x-axis*: points along main axis
- *Velocity* (signed speed): s
- *Steering Angle*: ϕ
- How does the car move?
 - In a small time interval Δt , the car must move approx in the direction of the rear wheels
 - In the limit, as $\Delta t \rightarrow 0$, then $dy/dx = \tan \theta$, i.e.,
 - $\dot{x} \sin \theta + \dot{y} \cos \theta = 0$
 - Solution is of the form
 - $\dot{x} = s \cos \theta$ and $\dot{y} = s \sin \theta$
 - What about $\dot{\theta}$?



Express car motions as a set of differential equations

$$\dot{x} = f_1(x, y, \theta, s, \phi)$$

$$\dot{y} = f_2(x, y, \theta, s, \phi)$$

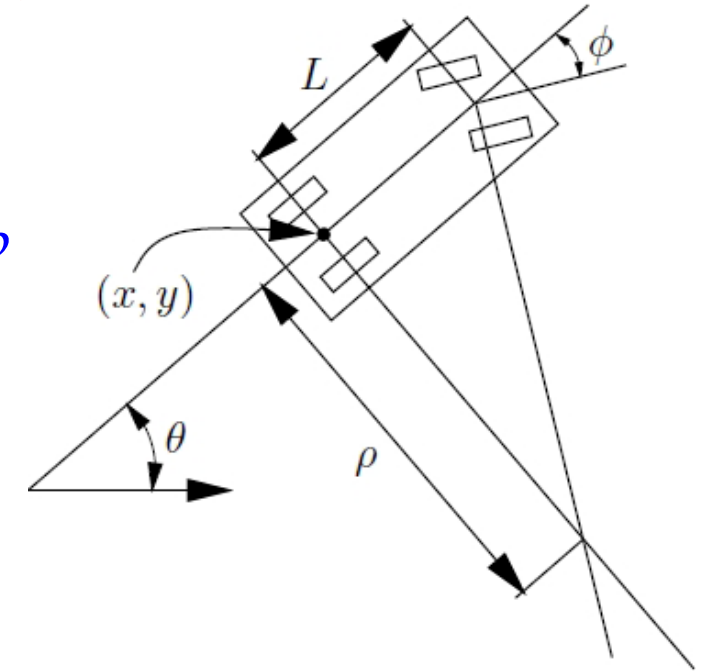
$$\dot{\theta} = f_3(x, y, \theta, s, \phi)$$



- Introduction
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Kinematics for Wheeled Systems – A Simple Car

- w : distance travelled by the car
- $dw = \rho d\theta$
 - If the steering angle is fixed at ϕ , the car travels in circular motion, in which the radius of the circle is ρ
- $\rho = L / \tan \phi$
where L is the distance from front to rear axles
- Therefore
$$d\theta = \frac{\tan \phi}{L} dw = \frac{\tan \phi}{L} s \Rightarrow \dot{\theta} = \frac{s}{L} \tan \phi$$
- How should we control the car?
 - Setting the speed s , i.e., $u_s = s$
 - Setting the steering angle, i.e., $u_\phi = \phi$
- Putting it all together
 - Input controls:
 u_s (speed) and u_ϕ (steering angle)
 - Equations of Motions



$$\dot{x} = u_s \cos \theta$$

$$\dot{y} = u_s \sin \theta$$

$$\dot{\theta} = \frac{u_s}{L} \tan u_\phi$$



- Introduction
- Coordinate Systems
- Kinematic Models
- Constraints
- Applications
 - A Simple Car
 - Wheeled Robot
 - Unicycle

Kinematics for Wheeled Systems – Unicycle

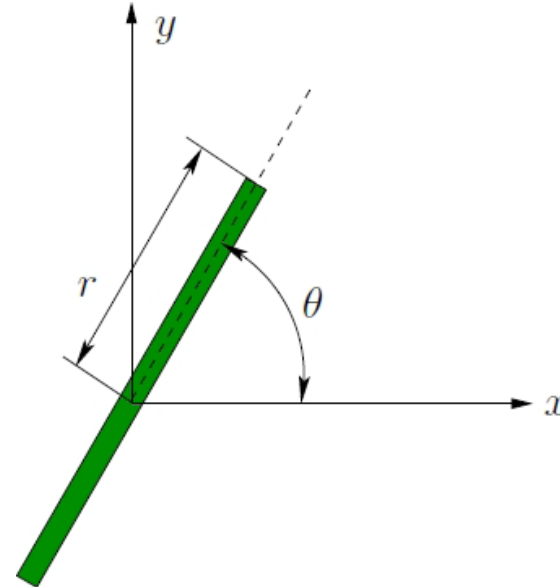
- *Input controls* rider can set the pedalling speed and the orientation of the wheel with respect to the z-axis

r : wheel radius

σ : pedalling angular velocity

$s = r \sigma$: speed of unicycle

ω : rotational velocity in the xy plane



Equations of Motions

$$\dot{x} = u_{\sigma} r \cos \theta$$

$$\dot{y} = u_{\sigma} r \sin \theta$$

$$\dot{\theta} = u_{\omega}$$

What about acceleration?

